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SuperHive: Permitting AI

In March 2025, ai71, an Abu Dhabi-based company working to speed up building permits in the UAE and beyond faced a permissions question of its own: Should it give itself the greenlight on SuperHive, its AI construction intelligence product, before the technology was 100 percent accurate?

SuperHive, known in its early days as Cons71, was conceived as an “AI-powered Real Estate Assistant” with uses for firms and for governments. The feature that was at least 90 percent ready by Spring 2025 was its automated compliance checking. The technology could automatically scan site and building designs to compare them against local codes and regulations. All over the world, building codes and the processes for navigating them could often be labyrinth. Governments, the thinking went, could use these tools to help in permit review. Firms could use them in advance to aid in shaping permit applications. The potential of SuperHive was to reduce permitting times from months to minutes; and beyond that to transform everything from design, to modeling, to space optimization, to engineering and mechanical integration.

The compliance tool was just one feature set of SuperHive, which was just one early product of ai71. The company was also underway with other tools that had government and private use cases, in healthcare, for example, and in law. And products were only one aspect of ai71’s developing business model. The company provided access to large language models (LLMs) via their platform. And they were building out advisory services to try to help government and firms make productive use of AI.

More was perhaps resting on the launch of the construction product than just its success or failure. ai71 was stood up by VentureOne, an arm of the UAE’s Advanced Technology Research Council, in large measure to commercialize the United Arab Emirates’s (UAE) family of LLMs, Falcon. These open-source models, developed by the UAE’s Technology Innovation Institute (TII), had landed with fanfare in 2023, when they topped some global rankings. Now, the company—and indeed the models—were part of a larger national AI strategy to diversify the UAE economy and strengthen its AI sovereignty.

H.E. Faisal Al Bannai, ai71’s chairman, had launched the company in 2023, “to bring the power of open source LLM models to the world through transformative AI-powered enterprise solutions.”¹ Given the backlog of building permits around the world, not to mention housing shortages, ai71’s technology seemed potentially transformational. But like many AI tools at the time, it wasn’t 100 percent reliable. Amidst a race for AI sovereignty and profitability, for talent and for market share, the company would have to decide when it came to their AI, how good was good enough?

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UAE + AI

ai71 and SuperHive were new elements of an expansive vision for AI in the UAE. In 2017, the country launched its National Artificial Intelligence Strategy 2031 as part of its ambition to transition into a knowledge-based economy.^{2,3} The UAE, home to roughly 10 million people, was founded by the late sheikh Zayed bin Sultan Al Nahyan in 1971 as a federation of seven Emirates with Abu Dhabi as the capital.⁴ A member of the organization of petroleum exporting countries (OPEC), oil had played a crucial role in the country's economy since its inception.^{5,6} Particularly since the early 2000s, the country had made efforts to diversify its economy away from oil.⁷ To that end, the UAE had launched several national strategies designed to grow emerging sectors.⁸ AI was one of them. By 2024, non-oil activities contributed to 74% of GDP (compared to 69% in 2015).^{9,10} IBM forecasted AI to contribute \$96 billion to UAE's GDP (14% of total) by 2030.¹¹ (See **Exhibit 1** for an overview of the AI strategy.)

The UAE was perhaps well positioned to succeed in AI.¹² It had wealth to purchase the hardware necessary.¹³ It had electricity to power it all, given its oil and renewable energy (particularly year-round solar). Dr. Hakim Hacid, chief researcher at TII, said, "We don't have a shortage of resources, and the AI sector is getting everything it needs to progress and achieve something great." The country was also working to attract businesses and talent alike, promising high standards of living, no income tax, and more.¹⁴

Additionally, the UAE government had been characterized as tech-savvy and had positioned itself as such.¹⁵ It boasted a highly digitized government with plans to be the world's first fully AI-native government by 2027.¹⁶ It offered AI businesses access to large sets of anonymized data on its residents, including those from hospitals and state-backed entities.¹⁷ At a virtual event in 2024, Sam Altman, co-founder and CEO of OpenAI, highlighted the UAE's potential as a "regulatory sandbox" for AI experimentation.¹⁸ Mehdi Ghissassi, chief product and technology officer at ai71, commented, "In general, governments are resistant to using AI. But that's not true for the UAE, they're actually more keen to try it out than the private sector."

Since the announcement of the national AI strategy, the UAE had made headlines with several multi-million and billion-dollar investments in infrastructure, developing technologies, and even in becoming a manufacturer of semiconductors used in chips critical for AI.^{19,20} Dr. Najwa Aaraj, CEO at TII, observed, "There is a very big commitment of the country overall to AI."

The investments were not restricted to the UAE. In 2022, G42, the Abu-Dhabi based technology holding company, launched a \$10 billion fund to "accelerate the pace of global innovation" by functioning as a private equity investor focused on late-stage investments in high-growth regions.²¹ It had the world's largest AI-dedicated supercomputer (located in the U.S.), built by American firm Cerebras.²²

Many of the investments and partnerships were with American entities and formed the basis of growing AI-focused ties between the UAE and U.S.²³ G42 had partnerships with several American firms including OpenAI, IBM, NVIDIA, and others.²⁴ In April 2024, Microsoft announced plans to invest \$1.5 billion in Abu Dhabi to help create a skilled AI workforce in the UAE and pave the way for collaborations on AI infrastructure in the Middle East, Central Asia and Africa.²⁵ The U.S. rolled back its restrictions on the amount of NVIDIA-class chips the UAE could import in May of 2025.²⁶

A Growing AI Ecosystem

The national AI strategy helped create an AI ecosystem in the UAE and Abu Dhabi in particular. The multiplying entities including startups, educational and research institutions, investment funds,

and more. Key players included new AI-focused regulatory authorities and other entities dedicated towards developing and commercializing AI technologies like G42 and the Advanced Technology Research Council (ATRC).

The ATRC, founded in 2020 was responsible for defining Abu Dhabi's technology research strategy; consolidating and facilitating funding; and policy making.²⁷ Its executive board was made of seven of the country's most influential figures and was chaired by Abu Dhabi's crown prince sheikh Khaled bin Mohamed bin Zayed Al Nahyan.²⁸ The ATRC carried out its efforts through three entities; Aspire, TII, and VentureOne. Aspire focused on developing technology talent. TII, meanwhile, was responsible for researching and developing the technologies. (See **Exhibit 2** for TII research areas.) Its most well-known was Falcon, the LLM, named after the country's national animal. VentureOne was set up to commercialize those technologies by refining them, converting them into products, and then selling them both locally and internationally.

The Abu Dhabi Digital Government Strategy

AI efforts in Abu Dhabi extended into the administration of government. In January 2025, the Abu Dhabi government announced the Abu Dhabi Government Digital Strategy 2025-2027, with a goal to advance "the emirate's transformation towards an AI-powered government."²⁹ That meant shifting all of the government's operations onto the UAE's sovereign cloud, and digitizing and automating all government processes.³⁰

The plan earmarked roughly \$3.5 billion to be deployed by the Department of Government Enablement (DGE).³¹ Reda Nidhakou, CEO of VentureOne, commented, "There's a real drive to adopt AI in the UAE. DGE, for example, has put a team of AI experts together to enable all the entities within the government to use AI. We have been overwhelmed with end user demand in the early days of ai71." The DGE worked closely with many of Abu Dhabi's entities; it had a preexisting responsibility to create government-wide policies on three key domains: human resources, digital and cybersecurity, and procurement.^{32,33} It had already spent over a decade digitizing the government and its services to residents and businesses.³⁴

Much of that work had come to fruition in the TAMM platform, where AI would also now be brought to bear.³⁵ TAMM (an Arabic term which meant "consider it done") was an application that could be downloaded onto a user's smartphone.³⁶ It catered to Abu Dhabi's 2.5 million residents and offered them the ability to carry out 1,000 government services (e.g. paying speeding tickets and utility bills, registering a marriage, or even applying for a permit for large gatherings like a private party).^{37,38} In 2025, users could, for example, ask the TAMM-AI assistant how to complete certain requests like renewing their vehicle registration.³⁹ They could ask the chatbot to do the whole process for them. It could even obtain car insurance quotes and select one on the users' behalf in a matter of seconds.⁴⁰

Other users and uses emerged as well. The Abu Dhabi Department of Municipalities and Transport, for example, announced in 2024 several partnerships to obtain AI-powered customized solutions in the mobility sector to better manage traffic, deploy charging infrastructure for electronic vehicles, enhance transport safety and more.⁴¹ The Abu Dhabi National Oil Company announced plans to use agentic AI to slash the time taken for seismic surveys from months to days and boost the accuracy of oil production forecasts by 90%.⁴² The Abu Dhabi Judicial Department launched an AI-powered system available 24/7 to automatically reduce traffic fines without a court's involvement.⁴³

AI Sovereignty

The investments in AI were also meant to help the UAE go beyond government efficiency; they were part of an effort at “AI sovereignty”.

The notion had many definitions. NVIDIA characterized sovereign AI as “a nation’s capabilities to produce artificial intelligence using its own infrastructure, data, workforce and business networks.”⁴⁴ That conception could be expanded to include other aspects such as regulations and legislation around AI, strategic investments in research, and building an AI ecosystem.⁴⁵ AI sovereignty was said to be potentially and particularly valuable considering AI’s growing role in military, society, and politics.⁴⁶ Jensen Huang, NVIDIA’s founder and CEO had gone so far as to assert that, “Every country needs to own the production of their intelligence.”⁴⁷

AI sovereignty was an evolution of an older concept: data sovereignty.⁴⁸ Long before LLMs, governments had been making choices about whether to store data on their servers that they owned and were located on their own soil or in their own sovereign cloud. AI sovereignty went beyond data sovereignty and included the AI, including LLMs, which parsed or were built from that data.⁴⁹ Owning the AI models was said to help governments avoid the risk of relying on foreign entities for access.⁵⁰

NVIDIA’s leadership and UAE’s seemed to share a perspective on this. In 2024, at a fireside chat at the World Government Summit held in Dubai, Huang commented on sovereign AI, “It codifies your culture, your society’s intelligence, your common sense, your history.”⁵¹ Omar Al Olama, the UAE’s minister of AI, responded, “We completely subscribe to that vision...That’s why the UAE is moving aggressively on creating large language models and mobilizing compute.”⁵²

Models built by local teams might also play a role in protecting and promoting national language, culture and values.⁵³ Dr. Aaraj observed, “Having your own models helps get rid of any biases towards the region that may exist in other models.” A Deloitte report on sovereign LLMs elaborated, “the reality is that many platforms are often biased toward their country of origin across several dimensions, including legal frameworks, ethical and equity frameworks, language, and culture.”⁵⁴ Several LLMs had been developed in the UAE, including Jais, Nanda, and Sherkala by G42, and Falcon by the ATRC’s TII.

Even if desired, achieving AI sovereignty had its challenges. It was costly in terms of energy and cutting-edge hardware.⁵⁵ It required tech talent and developed regulatory and governance frameworks.⁵⁶ Much of the leading AI technology was concentrated in a small number of private tech firms.⁵⁷ Moreover, as governments invested in developing their AI, they needed to navigate decisions on whether to compete or collaborate (or both) with the private sector.⁵⁸ Some governments settled for having the ability to control how models were used as opposed to fully controlling or providing the technology themselves.⁵⁹

In 2025, over 60 countries had national AI strategies.⁶⁰ Saudi Arabia, which neighbored the UAE, had an AI strategy focused on developing talent and regulation, research and collaboration, developing an AI ecosystem, and providing and attracting funding towards those efforts.⁶¹ India espoused AI as a tool for inclusive growth and sought to position itself as a testing ground for global AI innovations catered to emerging markets.⁶² Stanford University’s Global AI Power Rankings took into consideration 42 indicators from research papers to patents to investment and more and in 2024 ranked the top five AI-nations as the U.S., China, the U.K., India, and the UAE.⁶³

Falcon

Falcon had helped put the UAE on the AI map. When Falcon 40b launched in March 2023, it topped leaderboards on Hugging Face and was reported to outperform several comparable models, including ChatGPT-3.⁶⁴ Dr. Ebtesam Almazroui, a director at Abu Dhabi's AI and Digital Security Research Center remarked in a statement that Falcon 40b "[matched] the performance of Chinchilla (from DeepMind by Google) and PaLM-62B (also from Google), at considerably lower training costs in comparison."⁶⁵ It also required less compute power than similar LLMs like Meta's Llama.⁶⁶ One AI researcher at Hugging Face said in a statement in *Time Magazine*, "The UAE was not well known, before, for training models... But then, by more or less the next day, we knew that they can train models, they open-source their models, they publish research around it, which benefits all."⁶⁷ Nidhakou expressed even the team's wonder at all this, "We knew it would do well, but we were honestly pleasantly surprised with both its performance and traction."

Developing Falcon had not been cheap. The 384 A100 NVIDIA chips used to train Falcon 40b had cost \$10,000 each; it was not unusual for LLMs to cost \$20 million or more to develop.^{68,69} On top of that was the cost of the compute to support its daily use. Despite the costs, TII would earn no revenue from Falcon; it was open-source and free to the world.^{70,71}

TII's leadership had felt open-sourcing Falcon was the right call. Nidhakou remarked, "We knew 40b was only the beginning of our journey. We were confident that the value it would bring in was worth making it open source. It at least meant Falcon 40b could be tested by countless people around the world." Dr. Hacid added, "It made us part of a very active and collaborative global community. Developers in the community let us know how the model can be improved and come up with use cases we didn't think of." Developers sometimes preferred open-source LLMs since they could be fine-tuned, and there were several available.⁷² Some open-source LLMs had restrictions regarding commercial use, but Falcon's leadership elected not to put any in place.

Being open source, the team believed, also aligned with UAE's vision to make an impact, which in turn would attract world-class talent. Omar Al Olama, UAE's first AI minister said, "If you bring a person who is the top scientist in this field, and you [give] two options: either work for-profit, closed-source, and just benefit yourself, or work on something that's open-source that will really change the lives of people, what would you decide?"⁷³

Falcon Models

Between 2023 and 2025, TII released several more Falcon models, big and small. (See **Exhibit 3** for model timeline and **Exhibit 4** for benchmarking). The largest was Falcon 180b. It was a "super-powerful" model with 180 billion parameters and, according to TII, once again beat competitors from Meta and Google and rivaled OpenAI's closed source ChatGPT-4.⁷⁴

Although TII had built leading models, it was a challenge getting people to use them. Dr. Hacid explained one of the reasons, "The AI we built performed really well in tests, but it wasn't necessarily good when it came to *actual* usage by developers."

So, newer models, like Falcon 3, were built with adoption in mind. Released in December 2024, Falcon 3 was a much smaller LLM than its predecessors at under 10 billion parameters. Dr. Aaraj said, "Bigger models are valuable and can be more capable than smaller models, but the adoption can be limited by the availability of compute power of users." Smaller models were easier to deploy on cheaper hardware; Falcon 3 could be run on a laptop. It was not only smaller, but it was also more

efficient. Falcon 3 became TII's flagship model. Dr. Hacid said, "It's really different from everything we've built in the past in term of data, how we trained it, and even in terms of performance."

Building a smaller model seemed to much of the team an avenue worth pursuing. Dr. Hacid said, "It's great to have leaderboard-topping LLMs, but at the end of the day what's most important is that these models lend themselves well to the community's needs and helps solve real world problems." And the open-source community seemed to prefer smaller models. He said, "We will of course continue to work on large models, too, but smaller models will be part of the focus; why should we put so much effort into building a double-decker bus if all the community really wants is a four-door sedan?"

There were other obstacles that were slowing adoption. Dr. Hacid explained, "It's not enough to have our models available to the community to find and download. We need partners to help us push these models to potential users. They can help by repackaging the models into different solutions for specific needs and then sell it out to their customers." The team reached out to hyperscalers, such as Amazon Web Services, and commercial entities, including, ai71.

Falcon's Competition and Peers

Falcon's models evolved in an ever-changing landscape alongside several other open-source and closed-source LLMs from around the world. The LLM market was estimated at \$6 billion in 2024 and projected to grow to \$36 billion by 2030.⁷⁵

Top performing LLMs had generally been closed-source. They were developed and operated by private companies and included Google's Gemini, OpenAI's ChatGPT, and xAI's Grok among others.⁷⁶ Those companies had spent hundreds of millions of dollars on developing the LLMs. Nidhakou commented, "Falcon doesn't really compete with OpenAI. It doesn't make sense to compare Falcon with ChatGPT unless it becomes open source."

But open-source models were proliferating and performing. Some of these were produced by private entities, while others by governments around the world. Leading open-source LLMs included Google's Gemma, Meta's Llama and others. DeepSeek became one of the most prominent open-source LLMs when it burst onto the scene in January 2025.⁷⁷ It caused a stir as, not only was it developed by a lesser-known Chinese startup, but it was also reported to be as capable as leading closed-source LLMs, built at a fraction of the cost⁷⁸ and despite U.S. restrictions on selling NVIDIA chips to China.⁷⁹

Reactions to DeepSeek's model varied. Analysts expected investors would rethink betting on closed-source models to win the AI race.⁸⁰ NVIDIA's stock dropped by \$600 billion as investors wondered if demand expectations were still relevant, though reached new highs six months later.⁸¹ Asked whether DeepSeek was "competition" for Falcon, Dr. Hacid commented, "We've always believed open-source models would eventually be on par with closed-source ones. We were excited to see DeepSeek prove it. It's really helped encourage the whole open-source community to push the boundaries further."

A survey conducted by the prominent US-based venture capital firm Andreessen Horowitz suggested open-source LLMs were becoming more popular among enterprises, and that nearly half of the respondents would switch to using open-source LLMs if they matched the performance of closed-source LLMs.⁸²

ai71

As the world became more familiar with Falcon, the team saw opportunities to try to boost its adoption and monetize it. “Getting Falcon into people’s hands was one thing. Making sure it had a real impact was another,” Nidhakou said. “We wanted to solve big problems, have a meaningful impact to society, and generate an economic benefit for us.”

The first avenue was VentureOne. Founded in 2022, its handful of specialized subsidiaries served several sectors namely autonomous vehicles, climate technology, cybersecurity, defense, logistics, and others.⁸³ Dr. Hacid said, “The idea was to have a dedicated team of experts focused on repackaging Falcon into different solutions and then selling it, rather than having TII’s manpower also working on commercialization.”

ai71 was one of the subsidiaries. It operated in a competitive environment with other sovereign and private developers of AI for governments and consultancies. ai71 promoted its access to large, anonymized datasets from the UAE and decentralized data ownership.⁸⁴ The latter meant clients would retain ownership of their data. The team felt this was a “game changer” for entities focused on privacy.⁸⁵

Products

For ai71, the first piece of the puzzle was identifying which sectors to target. The team began searching for difficult problems they could solve in high-GDP sectors with little competition. Initially, they focused on the medical, educational, and legal sectors.⁸⁶ Each of the early products were built using Falcon LLMs initially but not exclusively and, by 2025, were still in development.

Cons71 was envisioned as an AI-powered construction solution that would help automate the issuance of building permits by checking designs against relevant rules and regulations; an otherwise mainly manual process. Later it would be expanded and renamed SuperHive.

Razi71 was developed as an AI-enabled healthcare solution to review and summarize medical literature, transcribe case notes, cross-check medications, and more. Its purpose was to free up healthcare professionals so they could spend more time engaging with patients.

Ask71 was conceived as an AI-powered assistant that could use knowledge across an organization to generate insights and improve efficiency. Ask71, also known as AskGOV, could for example help answer an employee’s questions on specific human resources policies like how many leave days they were entitled to and then apply for their holiday. It was also intended to function as an assistant to business leaders, helping them leverage data and insights across their organizations to answer difficult questions.

Platform

In addition to these products, one of ai71’s most recent offerings was the **ai71 API Hub**. It was launched in 2024 as a platform that allowed pay-as-you-go access to Falcon and other AI models. The idea was to make it easier for users – and not just business or government clients – to use big models like Falcon 180b without needing costly infrastructure.

ai71 planned to start with Abu Dhabi government entities and then sell products to customers all over the world. Chiara Marcati, ai71’s chief AI advisory and business officer, said, “When we develop a new product, we first deploy it with a small group of trusted design partners. These organizations

are close enough to give candid feedback and patient enough to iterate with us. This creates a fast learning loop before we scale the product more broadly.”

Services

Selling AI as a product didn’t prove seamless. While still developing them, the team at ai71 met with various potential customers and stakeholders to hone the product design. “We started engaging with potential clients and found out that, although they were eager to use Falcon and AI in general, they didn’t know where to start,” said Nidhakou.

They formed an advisory arm at ai71. Nidhakou explained, “We realized that different organizations had differing abilities to adopt AI, and that we owed it to the ecosystem to advise them on how to best deploy AI in their organizations.” The advisory arm would help clients identify ways they could benefit from AI in their organizations, develop the AI solutions, and help with implementation. “Our advisory engagements are not separate from our product journey. They allow us to understand, in real time, the operational problems organizations face and translate those directly into product capabilities.” Marcati added, “The entities we work with are also central to AI decision making across the UAE government, which gives us a uniquely fast feedback loop and access to high quality, real operational data to continuously improve our products while solving mission-critical problems. In addition, it allows us to get deep into the adoption challenges that enterprises are facing.”

One of those clients was the DGE, the entity responsible for TAMM. ai71 was working with them to add new dimensions to TAMM’s conversational abilities.⁸⁷ Initially, the plan was to use Falcon. However, TAMM’s chatbot was ultimately powered by Microsoft Azure and OpenAI. Nidhakou commented, “We recommend Falcon when it’s the right tool for the job, and sometimes it isn’t. It’s in the best interest of Falcon to tell clients if we think they’re better off using another LLM and then come back to them a few months later when Falcon is ready for their specific use case.”

SuperHive

By January 2025, several potential clients showed interest in the construction product, and the ai71 team began focusing their efforts on developing it further.

SuperHive was created to address the often lengthy process of obtaining a permit from government to construct or alter a building. The team at ai71 learned that the approval process for a building permit in Abu Dhabi took at least one month. “That was assuming it was a simple design, like a villa,” explained Klemensas Mecejus, head of advisory at ai71. “Designs for complex projects like skyscrapers would take even longer to approve.” Mecejus added, “An application would typically have 20 or so documents that needed to be assessed for compliance against hundreds of rules and regulations, and checking compliance with these rules was not necessarily straightforward.” Mecejus continued, “The regulations aren’t organized as a checklist. Rather they are often written in paragraphs of text. It’s not always easy to recognize the individual rules.” In addition, because there were so many rules to check, the designs for a building were reviewed by multiple people which stretched the process out further.

SuperHive was at first to be a tool that understood all of the government’s construction-related rules and regulations and could quickly assess to what extent the plans for a building were compliant. ai71 targeted stakeholders on both sides. Developers could test and adjust their designs before even submitting them to the government. Regulatory authorities could be relieved of or supplement the manual checking of designs.

The backlog of permitting across the world and the demand for housing and new construction broadly had drawn ai71's interest into this space and also others'. In the U.S., the startup CivCheck had launched an "AI plan review platform" they said was "capable of supporting every check required for building permits and planning approvals."⁸⁸ It was designed to "augment rather than replace staff, with every AI interpretation reviewed and approved by staff for the most accurate and trustworthy results."⁸⁹ CivCheck asserted they could reduce permit approval times by 80% or more.⁹⁰ Scale AI, which had started as data annotation company, but expanded quite substantially across AI products and services, was offering "agentic building permit validation" which promised to "automate the assessment of complex construction plans for compliance with building codes and safety standards."⁹¹

To flourish, SuperHive would need to succeed on a few fronts. For starters, the tool had to know all the relevant construction rules and regulations. It also had to be capable of receiving and processing several types of files and documents ranging from relatively simple Word and PDF files to complex AutoCAD files that showed two-dimensional (2D) blueprints of buildings, to even more complex Building Information Modeling (BIM) files with three-dimensional (3D) renderings of buildings. "And then," Mecejus added, "It needed to be able to analyze those files, assess how compliant the design was, and point out exactly where it failed to comply and why."

The team used the insights from their interactions with clients to build SuperHive. Mecejus explained, "We used a combination of LLMs and natural language processing to develop a classification algorithm we call 'RuleBuilder' to reorganize all the relevant regulations into a checklist." With the checklist in hand, the team needed to get SuperHive to understand building designs. For 2D, they used a computer vision algorithm which they had trained. For 3D, they used a simpler approach. Mecejus explained, "The 3D files always come from BIM, which means they have a standardized format. That made it relatively easy to create our own rules into the product on how to digest those files." The team then taught the model to assess compliance. Mecejus commented, "We set up proper computations between the algorithms so that a specific rule could be assessed against 2D and 3D files and pinpoint where there were issues in the design."

SuperHive employed a variety of AI models. Marcati said, "When it comes to AI models, there is no one-size-fits-all model. We use Falcon, Llama, ChatGPT, and others since they all excel at different things." The decision on what AI to use depended on the task at hand and other considerations. Marcati elaborated, "An important determining factor is to do with the client's limitations. Abu Dhabi's laws, for example, necessitate local hosting of AI servers, so we're restricted to Falcon, Prem or OpenAI."

SuperHive provided the user with a report on how compliant a design was and explained which regulations were not met and why. (See **Exhibit 5** for an example). The report also highlighted which parts of the drawings could not be assessed. This occasionally occurred if SuperHive was unable to understand a design or parts of it.

Evaluating SuperHive

By the spring of 2025, the ai71 team was gathering an early sense of the product's performance and its prospects.

They'd been accumulating a pipeline of clients for SuperHive, governments and otherwise. Marcati said, "Although SuperHive was born in construction, the underlying computer vision engine is modality agnostic." The idea was that the logic SuperHive used to assess compliance could be applied to other industries. Marcati said, "It interprets structured visual artifacts against regulatory standards.

Whether the input is an architectural drawing, an electrical plan, or another technical schematic, the logic remains the same.”

And they saw a roadmap for an expansive platform to other parts of the construction value chain that SuperHive could support. These included 1) investment and strategic planning; 2) design and planning; 3) construction; and 4) post-construction (i.e. operating, maintaining, and then demolishing the building). Marcati said, at first SuperHive “only addressed the design stage, but we want to make a tool that is a one-stop-shop for construction leaders. We want it to be the ERP equivalent for the construction industry; something seamlessly integrated with all the popular specialized tech in that field.” They’d begun developing the capabilities for “digital twins”.

They were seeing high accuracy rates. At their offices, when they measured their AI’s assessments vs. what the permitting rules dictated was allowable, they were finding 90% accuracy. Mecejus added some nuance to their evaluations, “Not all rules are created equally. Some regulations are more important than others. For example, ensuring that a structural column would be capable of holding up a ceiling was far more critical than making sure that electrical sockets were exactly the right number of inches away from each other.” Critical rules were identified and prioritized. Mecejus said, “Those had to be assessed perfectly, every time, no matter what.”

There were other measures that mattered as well. Like speed. By March 2025, their AI permitting tool was able to process drawings in a minute, substantially faster than humans, and the team were confident they could eventually get it to be instantaneous. Whether humans should be “in the loop” was up for debate in AI circles, but the team still recommended so. Marcati said, “Clients were fairly open to using the AI because the value it offered was clear, but they were apprehensive about totally letting go of control.” Insisting on having a human review SuperHive’s output meant that it was likely the approval process would take significantly longer than only a minute. “But,” Mecejus said, “that meant it might take a day or two to review a design. That was still better than months. Plus, it was only one person in the review process. The others could be freed up for other tasks.”

One measure that seemed of growing importance to clients was compute demands. Mecejus said, “I began my career as a machine learning engineer in 2012. Back then, people never wondered how large the code was for a certain application. But clients are prioritizing this now.” So, ai71’s project milestones include getting the AI to reach a certain level of optimization. “That way,” Marcati explained, “clients can contain the cost of AI by avoiding costly upgrades to their AI infrastructure in the future.”

Certainly the “intelligent construction suite” the team were building SuperHive into had to deliver for the stakeholders they were targeting. For governments, they were promising streamlined monitoring and automated regulatory checks. For developers and investors, they were offering accelerated timelines and maximized project returns. For construction contractors, they were pledging help staying on schedule. And for consultants and engineers, they were promoting faster and fewer revisions with confidence in code compliance.

The question was whether to give their own permitting application the go-ahead, and how would they know?

Exhibit 1 Overview of the UAE National AI Strategy

Objective	Planned Actions
Build reputation as an AI destination	To create a new AI certification system: UAE Seal of Approval To become a global AI Hub by hosting conferences, and becoming a center for AI startups
Develop competitive assets in priority sectors	To provide government incentives and advice for local firms To encourage global AI leaders to locate in UAE
Develop an active AI ecosystem	To provide government incentives for local and international firms, funding to startups and set up accelerators To develop a coordinated network of researchers, industry experts, and regulators To attract FDI
Adopt AI across customer services in government	To solicit ideas from government, universities, and private sector to solve key societal problems To set up a council to identify where and how AI can be used in government
Attract and develop talent	To provide AI courses dedicated to government staff and incorporate AI into education
World-leading research capability to collaborate with key industries	To create world class institutions to aggregate local and international expertise (e.g. National Virtual AI Institute) To create the Key Thinkers Program to attract leading AI minds To create an open-access digital AI-library of research
Become test bed for AI by providing data and infrastructure	To leverage UAE's diverse population – a unique source of data, particularly useful for health research To develop a secure data infrastructure that is easily accessible
Focus on governance and regulation	To take a leading role in governance To collaborate with other governments and thought leaders To run pilots to test theories

Source: Created by the case writers using information from “UAE National Strategy for Artificial Intelligence 2031,” UAE Minister of State for AI, Digital Economy, & Remote Work Applications, <https://ai.gov.ae/strategy/#opportunities1>, accessed May 21, 2025.

Exhibit 2 TII's Research Centers

Advanced Materials
AI and Digital Science
Autonomous Robotics
Biotechnology
Cryptography
Directed Energy
Propulsion and Space
Quantum
Renewable and Sustainable Energy

Source: Prepared by the case writers using company documents.

Exhibit 3 Timeline of Falcon Models

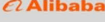
2023
Falcon 40B
Falcon 7B
Falcon 180B

2024
Falcon Mamba 7B
Falcon-E (Edge Series)
Falcon 2 – 11B
Falcon 2 – 11B VLM
Falcon 3 (family)

2025
Falcon Arabic
Falcon-H1 (family)
Falcon H1R 7B

Source: Prepared by the case writers using company documents.

Exhibit 4 Falcon Benchmarking

	Base models	MUSR	BBH	MMLU_PRO	If_eval	GPQA	MATH	Avg
	Falcon3 10B	14.17	41.38	36	36.48	12.75	24.77	27.59
	Falcon3 7B	18.14	31.56	32.34	34.16	12.86	19.26	24.72
	Qwen2.5 7B	14.14	35.81	37.39	33.74	9.96	18.88	24.99
	Qwen2 7B	14.32	34.71	35.37	31.49	7.27	20.47	23.94
	Gemma2 9B	14.3	34.1	34.48	20.4	10.51	13.14	21.15
	Mistral-nemo-base-2407 (12B)	6.52	29.37	27.46	16.3	5.82	5.36	15.14
	Llama3.1 8B	8.98	25.29	24.95	12.7	6.15	5.14	13.87
	Instruct models	MUSR	BBH	MMLU_PRO	If_eval	GPQA	MATH	Avg
	Falcon3 10B Instruct	13.61	44.82	38.1	78.17	10.51	25.91	35.19
	Falcon3 7B Instruct	21.17	37.92	34.3	76.12	8.05	31.87	34.91
	Llama3.1 8B Instruct	8.41	29.89	30.68	78.56	2.35	19.34	28.2
	Qwen2.5 7B Instruct	8.45	34.89	36.52	75.85	5.48	0	26.87
	Qwen2 7B Instruct	7.37	37.81	31.64	56.79	6.38	9.44	24.9
	Mistral-Nemo-Instruct-2407 (12B)	8.48	29.68	27.97	63.8	5.37	6.5	23.63
	Gemma2 9B It	13.77	21.62	20.48	50.1	2.68	6.72	19.23

Source: Company documents. <https://falconllm.tii.ae/falcon3/index.html>, accessed January 2026.

Note: “MuSR (Multistep Soft Reasoning) is a benchmark for evaluating language models on multistep soft reasoning tasks specified in natural language narratives... The benchmark tests chain-of-thought reasoning capabilities across domains involving commonsense reasoning about physical and social situations.”

“Big-Bench Hard (BBH) is a suite of 23 challenging tasks selected from BIG-Bench for which prior language model evaluations did not outperform the average human-rater... focuses on evaluating complex reasoning skills including temporal understanding, spatial reasoning, causal understanding, and deductive logical reasoning.”

“MMLU-RPO (Measuring Massive Multitask Language Understanding PRO) extends MMLU by expanding multiple-choice options from 4 to 10, eliminating trivial questions, and focusing on reasoning-intensive tasks. Features over 12,000 curated questions across 14 domains and causes a 16-33% accuracy drop compared to original MMLU.”

“IFEVAL (Instruction-Following Evaluation) benchmark is for large language models, focusing on verifiable instructions with 25 types of instructions and around 500 prompts containing one or more verifiable constraints”

“GPQA (Graduate-Level Google Proof Q&A) is a challenging dataset of 448 multiple-choice questions written by domain experts in biology, physics, and chemistry. Questions are Google-proof and extremely difficult, with PhD experts reaching 65% accuracy.”

“MATH dataset contains 12,500 challenging competition mathematics problems from AMC 10, AMC 12, AIME, and other mathematics competitions. Each problem includes full step-by-step solutions and spans multiple difficulty levels (1-5) across seven mathematical subjects.”

Instruct models have been fine-tuned for following instructions and task completion.

Benchmark explanations from llm-stats.com, <https://llm-stats.com/benchmarks>, accessed January 2026.

Exhibit 5 Sample SuperHive Output

The residential plot contains the following components: Residential villa, Appendices of Services, Hospitality Annex, Sports supplement, Swimming pool, Garage, Electricity room, transformers, and sub-electricity, Pool pump room, Site formatting elements, External fences and entrances for individuals and cars, Light shades and parking awnings.	Compliant
All services on the Roof floor & Top Roof floor must be concealed from view from all angles.	Not Compliant
Maximum plot coverage for single unit detached villa should not exceed 70% from the plot area, inclusive of the residential villa and all other structures. Aesthetic protrusions not exceeding 50 cm are exempt.	Compliant

Source: Company documents.

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